Pankaj Karamchandani

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PROFESSIONAL SUMMARY

Senior Full-Stack Engineer and **Technical Lead** with 8 years of experience architecting scalable **SaaS platforms** and cloud-native **distributed systems** using **Microservices** and **Microfrontend** architectures. Expert in **React**, **Node.js**, **Python**, **TypeScript, JavaScript** and **AWS**. Successfully led cross-functional teams of 5 engineers, achieving a **70 percent reduction in operational costs** through **GenAI and LLM integration** and a **60 percent increase** in **developer velocity**. Proven track record in System Design, System Architecture, TDD (Test Driven Development), CI/CD implementation, and delivering high-availability enterprise solutions for Media and Insurance domains.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3.
Frontend: React 18**,** Microfrontends (Module Federation), Next.js (SSR/ISR), Redux Toolkit, Context API, Web Vitals, Material-UI, Ant Design, Responsive Design.
Backend: Node.js, Express.js, Event-Driven Microservices, GraphQL, RESTful APIs, Serverless Architecture, WebSockets.
Cloud & DevOps: AWS (Lambda, S3, API Gateway, DynamoDB, SQS, Cognito, CDK, EC2, Batch), Docker, CI/CD Pipelines (Bitbucket Pipelines), IAC (Infrastructure as Code).
AI & GenAI: OpenAI (GPT-4), Anthropic (Claude), Gemini, LLM Integration, Prompt Engineering, AI Agents.
Databases: PostgreSQL, MongoDB, DynamoDB, Redis, NoSQL.
Testing & Tools: Jest, Storybook, Webpack 5, Turborepo, Lerna, Git, Jira, Postman, Swagger.
Core Competencies: Technical Leadership, System Design, Agile/Scrum, System Architecture, TDD (Test Driven Development), Microservices Development, Cloud-Native Architecture, Stakeholder Management, Mentoring, Performance Optimization, Project Delivery & Process Improvement.

PROFESSIONAL EXPERIENCE

MarketCast | Lead Software Engineer | Pune, India | Mar 2024 – Present

- Directing a cross functional team of 5 engineers to deliver scalable enterprise research solutions, overseeing architecture planning, code reviews, and sprint management.
- Automated the complete **media scene writing process** using **LLM Integration** to generate descriptions and survey questions this replaced manual video reviews, reducing operational costs by **60 to 70 percent** and increasing revenue throughput.
- Implemented **AI driven** response analysis algorithms achieving **93 percent accuracy**, significantly enhancing data quality control.
- Architected a **Python-based Scene Matching Library** using **perceptual hashing** and frame by frame comparison to automate video deduplication and content mapping.
- Modernized the **Scene Coding** workflow by eliminating legacy manual macros and standardizing the process, significantly reducing operational errors and overhead.
- Architected a dynamic **Survey Builder Application**, successfully reducing internal survey creation time by **90 percent** through modular design.
- Developed a **Predictive Modelling Tool** to forecast pre and post-exposure interest and conversion rates, this replaced manual Excel workflows with automated benchmarking, empowering the Operations team to deliver faster data-driven reports.
- Built a centralized **Infield Monitoring Tool** that streamlined data collection for field researchers, eliminating data aggregation delays and simplifying multi-source tracking.
- Designed an automated client media lifecycle management system on AWS, optimizing storage utilization and reducing streaming costs.
- Mentoring junior developers on **clean code** principles and driving the development of internal data analysis tools.

MarketCast | Senior Software Engineer | Pune, India | May 2022 – Mar 2024

- Spearheaded the modernization of legacy research monoliths by migrating to a **React, TypeScript** and **Microfrontend** architecture (using **Turborepo**), reducing data processing latency by **50 percent**.
- Developed a reusable, modular **UI Component Library**, ensuring design consistency and accelerating development velocity across 5+ internal applications.
- Built real-time analytics dashboards using **AWS Serverless (Lambda, API Gateway)**, achieving 99.9 percent system uptime.
- Optimized infrastructure as code using **AWS CDK**, reducing cloud infrastructure costs by **40 percent** through efficient resource allocation.
- Established robust testing protocols using **Jest**, achieving 85 percent code coverage and minimizing production defects.

Capita | Software Consultant | Pune, India | Mar 2021 – May 2022

- **Orchestrated** the Agile SDLC for an enterprise resource management platform, acting as the primary technical bridge between business stakeholders and engineering teams.
- **Architected** real-time, bi-directional collaboration features using **SignalR** and **Ant Design**, enabling synchronized workflows for concurrent users with sub-second latency.
- **Optimized** application performance by implementing lazy loading and code splitting strategies, achieving a **35 percent reduction** in initial load time.
- **Translated** complex business requirements into scalable technical specifications, ensuring **100 percent on-time delivery** of critical business objectives.

Accenture | Software Developer | Pune, India | May 2018 – Mar 2021

- **Developed** scalable pension and insurance applications using **React** and **Java**, supporting a high-volume user base of **10,000+**.
- **Revamped** Redux state management and component architecture, reducing technical debt and achieving **76 percent faster page load speeds**.
- **Engineered** a robust bulk data ingestion engine capable of validating and processing **50,000+ Excel records** with **99 percent accuracy**.
- **Optimized** backend data processing workflows, improving overall system throughput by **45 percent**.
- **Collaborated** across distributed global teams (3 time zones) to deliver enterprise-grade financial software within an Agile/Scrum environment.

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals - Cloud Architecture and Services

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TECHNICAL PROJECTS AND CONTRIBUTIONS

- **NPM Package Publisher (@pknodemicroservices/common):** Authored and published a reusable Microservices utility library to the public registry to standardize error handling and middlewares.

HONORS & AWARDS

- **Best Performer Award** – MarketCast (2022, 2024 and 2025)
- **"You Carried the Day" Award** – Capita (2022)
- **Notable Contribution Award** – Accenture (2020)

EDUCATION

JECRC Foundation | Bachelor of Technology, Computer Science & Engineering | Aug 2014 – Apr 2018

ADDITIONAL INFORMATION

- **Visa Status:** Active **US B1/B2 Visa** and **UK Visa**.